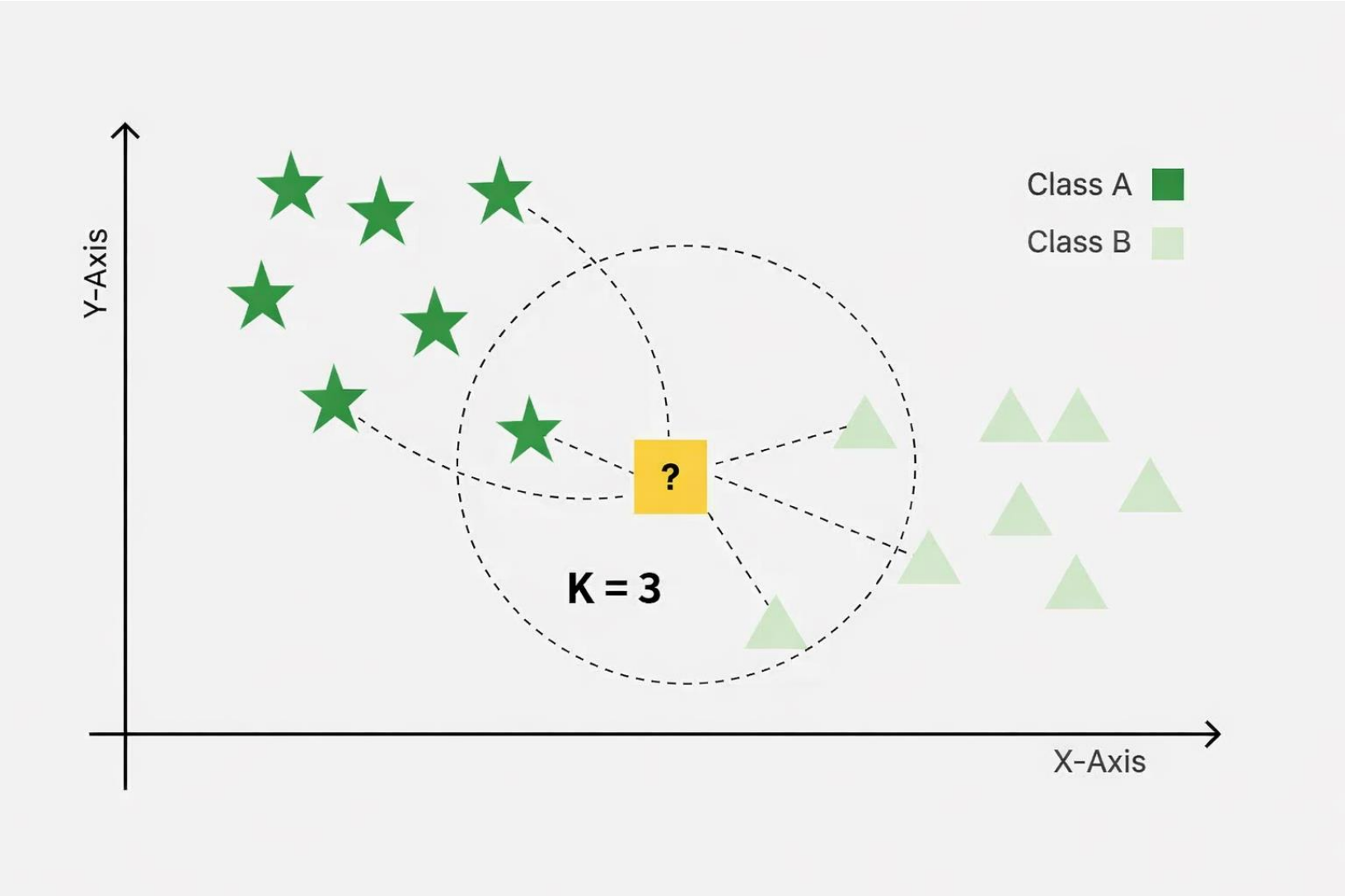
Module 7: Classification

Part 4: KNN

Wenting Xu

Iowa State University

K-nearest neighbor



K-nearest neighbor

- Takes a completely different approach than the other classifiers seen so far.
- KNN is completely non-parametric. Essentially, it makes **no underlying assumption** about the data or the shape of the decision boundary.
- We can expect it to dominate other classifiers when the decision boundary is **highly non-linear/irregular**.
- Drawback of KNN: you need a lot of data.

How it works ?

- Split data into training and test set.
- In order to classify an observation in the test set, I find the $\boldsymbol{K}$ **training observations** whose predictor values are closest to the test observation.
- Then the test observation is assigned to the class for which the **majority** of the K nearest training observations belong.

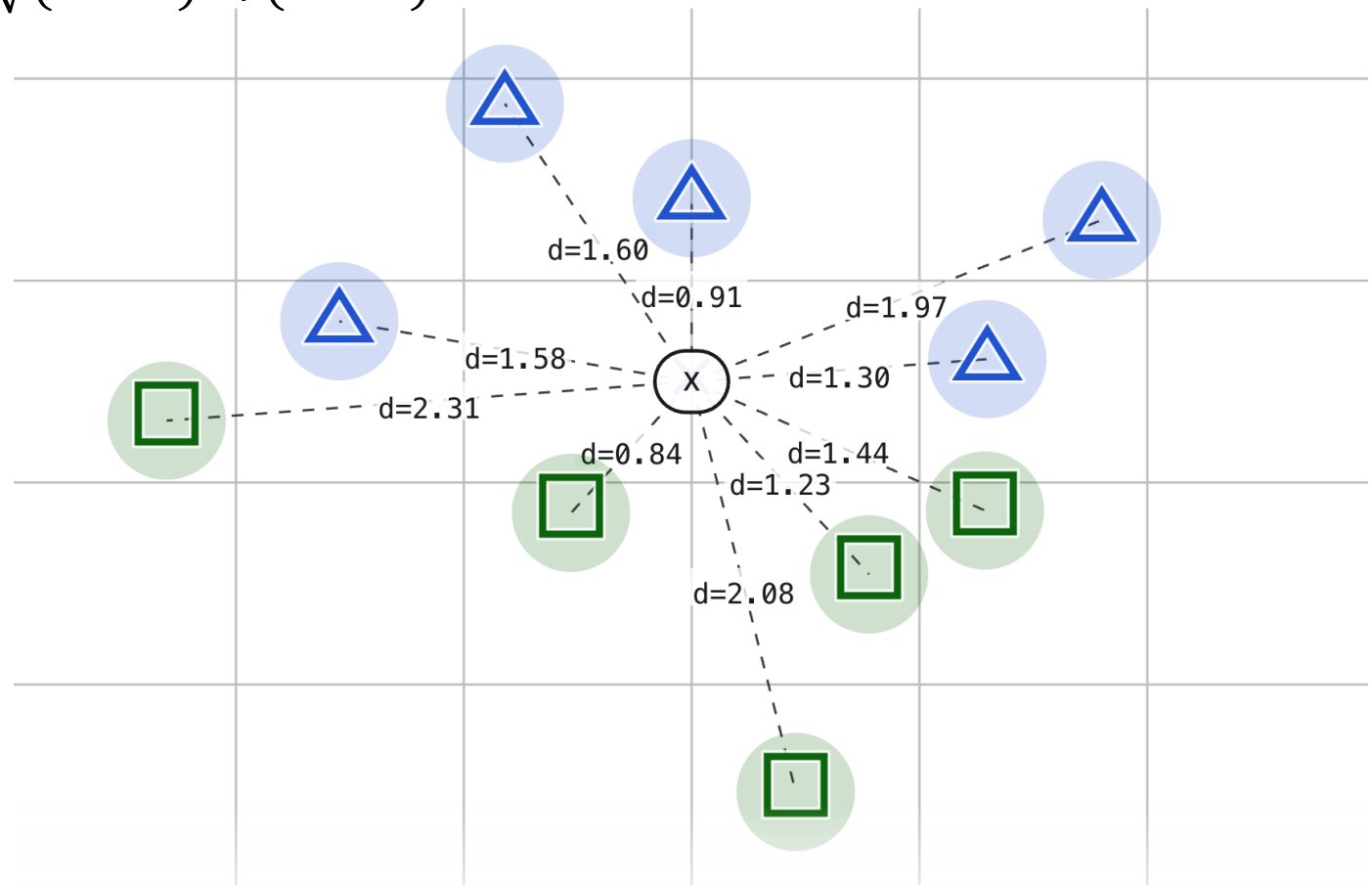
In-class activity

Step 1: Calculate distances

Euclidean distance:

Point A(2,3), Point B(5,7)

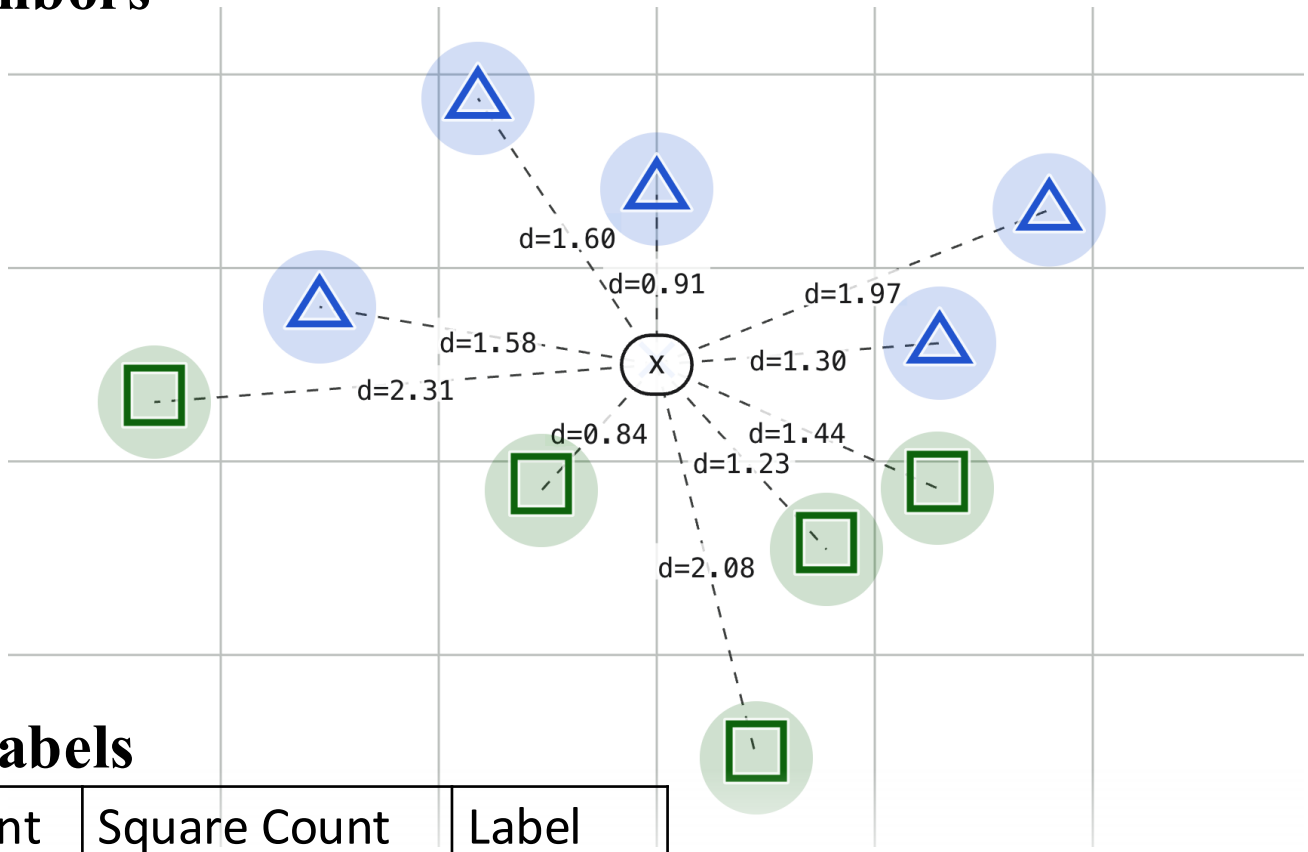
$$\mathbf{d} = \sqrt{(2-5)^2+(3-7)^2} = 5$$



In-class activity

Step 2: Find neighbors

1st NN	
2nd NN	
3rd NN	
4th NN	
5th NN	
6th NN	
7th NN	
....	



Step 3: Vote on labels

	Triangle Count	Square Count	Label
K = 1			
K = 3			
K = 5			
K = 7			

How can we handle ties for even k ?

KNN Demo

1. https://www.luseratech.com/knn-visual

2. http://vision.stanford.edu/teaching/cs231n-demos/knn/
   - Drag one point near the boundary. What happens to the decision boundary for different k?

How do we choose K ?

Choice of K affect bias-variance decomposition

Definition of ‘interest’

When the decision boundary is highly non-linear, KNN most closely mimics Bayes decision boundary.

How do we choose K ?

When the decision boundary is highly non-linear, KNN most closely mimics Bayes decision boundary.

Implementation

See R script: knn.R